

Introduction to SQL

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SQL = Structured Query Language

uses structured data

↳ used to manipulate, clean, process structured data.

SELECT statement $\Rightarrow$ first statement

↳ allows the specification of desired columns from the table

FROM statement

↳ allows us to specify the name of the table we are selecting the data from.

Students

ID	name	Subject
101	Roshi	Maths
102	Mark	Science
103	Elon	Physics
104	Jeff	Commerce
105	Kevin	Maths
106	Oprah	Science

$\Leftarrow$ SELECT *
FROM students;

• SELECT name
FROM students; =

name
Roshi
Mark
Elon
Jeff
Kevin
Oprah

SELECT DISTINCT subject
FROM students;

subject
maths
science
physics
commerce

SQL Operators

Arithmetic Operators ~ when applied a new column is created

Symbol	Description
+	Addition
-	Subtraction
*	Multiplication
/	Division

Products

ID	name	price	quantity
101	Laptop	1200	2
102	Mouse	250	4
103	Monitor	1000	2
104	keyboard	800	5

AS = Aliases

↳ to rename a column or table name.

-- Price hike by ₹100

• SELECT name,

price AS original-price

(price + 100) AS new-price

FROM products;

name	original price	new price
Laptop	1200	1300
Mouse	250	350
Monitor	1000	1100
keyboard	800	900

Calculate the revenue

```
SELECT name,  
        (price * quantity) AS revenue  
FROM products;
```

name	revenue
laptop	24000
Mouse	1000
Monitor	2000
Keyboard	4000

SQL Comparison Operators

Operator	Description
=	equals to
>	greater than
>=	greater than or equal to
<	less than
<=	less than or equal to
!=	not equal to
<>	not equal to

- used for data filtering. by using the WHERE statement.
- help us to create logical conditions for filtering the data.

```
SELECT column 1,  
        column 2,  
        column 3  
FROM table_name  
WHERE condition;
```



```
--  
SELECT name,  
        price,  
        quantity
```

```
FROM products
```

```
WHERE name = 'monitor';
```

→ affects the rows.

→ filtering rows.

name	price	quantity
monitor	1000	2

Data Filtering

↳ we use the WHERE clause

SQL Logical Operators

Operators

Description

AND

combines multiple conditions & all conditions must be met

OR

combines multiple conditions & one condition must be met

IN

allows us to specify values that we want on a specific column

NOT

gives us the opposite result

BETWEEN

allows us to filter data based on a specific range of value.


```
SELECT *  
FROM table_name  
WHERE condition;
```

condition $\Rightarrow$ we use operators to build conditions that help us to filter our data

Employees

ID	name	department	salary	age
1	Alice	IT	35000	26
2	Brian	Finance	45000	32
3	Carol	IT	50000	29
4	David	HR	30000	41
5	Emma	Finance	55000	35

① Write an SQL query to give the details of the employees in the IT department

```
SELECT *  
FROM Employees  
WHERE department = 'IT';
```

ID	name	de
----	------	----

② Write an SQL query to give me the name, emp-id, and salary of people earning more than £40 000

③ SQL query name of employees in the finance department & salary is over £50 000
(name, emp-id, department & salary)

```
2) SELECT name,  
           emp-id,  
           salary  
FROM employees  
WHERE salary > 40000;
```

```
3) SELECT name,  
           emp-id,  
           department,  
           salary  
FROM employees  
WHERE department = "finance" AND salary > 50 000;
```


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AND Statement

↳ used to combine multiple conditions

↳ All conditions must be met

```
SELECT column 1,  
        column 2,  
        column 3
```

```
FROM table name
```

```
WHERE condition 1 AND condition 2 AND condition 3... ;
```

- SQL query of employees in the finance department
whose salary is over 50000

```
SELECT emp_id  
       name,  
       department,  
       salary
```

```
FROM Employees
```

```
WHERE department = 'Finance' AND salary > 50000 ;
```

↓

emp_id	name	department	salary
5	Emma	Finance	55000

OR Statement

↳ to combine multiple conditions

↳ At least one condition must be met

- SQL query to give employee details of employees in IT or employees earning above 50 000

```
SELECT emp_id,  
       name,  
       department,  
       salary,  
       age
```

```
FROM employees
```

```
WHERE department = 'IT' OR salary > 50 000;
```

emp_id	name	department	salary	age
1	Alice	IT	35 000	25
3	Carol	IT	50 000	29
5	Emma	Finance	55 000	35

IN Statement (similar to OR statement)

↳ Allows us to filter the data base on specified values

```
SELECT column 1,  
       column 2,  
       column 3
```

```
FROM table_name
```

```
WHERE column_name IN ('Value 1', 'Value 2', 'Value 3')
```

• 'Value' is an entry in the column.

• Write a SQL query to find employees in IT or Finance.

```
SELECT
```

```
FROM
```

```
WHERE
```

name	dept
David	HR
Emma	Fin
Brian	Fin

```
BETWEEN
```

↳ Allows a range of values

```
SELECT
```

```
FROM
```

```
WHERE
```

min value & max value

• employees between 30 and 40

```
SELECT
```

```
FROM
```

```
WHERE
```


- Write a SQL Script to give employees IN HR AND finance.

```
SELECT name,
       department
FROM employees
WHERE department IN ('HR', 'Finance');
```



name	department
David	HR
Emma	Finance
Brian	Finance

BETWEEN Statement

↳ Allows data filtering based on a specific range of value within a given column

```
SELECT *
FROM table-name
WHERE column-name BETWEEN minvalue AND maxvalue.
```

^{min}value & ^{max}value are included

employees between the age of 20 to 30

```
SELECT name
       age
FROM employees
WHERE age BETWEEN 20 AND 30;
```

name	age
Alice	25
Carol	29